

An In-depth Technical Guide to the Physical and Chemical Properties of Neurine

Author: BenchChem Technical Support Team. **Date:** April 2026

Compound of Interest

Compound Name: *Neurine*
CAS No.: *463-88-7*
Cat. No.: *B1615796*

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For Researchers, Scientists, and Drug Development Professionals

Abstract

Neurine, or trimethylvinylammonium hydroxide, is a quaternary ammonium compound with significant toxicological properties. Formed from the decomposition of choline, it is often found in putrefying biological tissues. This technical guide provides a comprehensive overview of the physical and chemical properties of **neurine**, its synthesis and characterization, and its established neurotoxic effects. The information is intended to serve as a foundational resource for researchers in toxicology, neuroscience, and drug development.

Physical and Chemical Properties

Neurine is a poisonous, syrupy liquid with a characteristic fishy odor.^[1] It is soluble in water and unstable, readily decomposing to form trimethylamine.^[1] While a precise boiling point is not readily available in the literature, it is known to be a relatively non-volatile quaternary ammonium hydroxide.

Table 1: Physical and Chemical Properties of **Neurine**

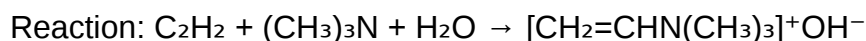
Property	Value	Reference(s)
IUPAC Name	Ethenyl(trimethyl)azanium hydroxide	[2]
Synonyms	Trimethylvinylammonium hydroxide, N,N,N-Trimethylethenaminium hydroxide	[1][2]
CAS Number	463-88-7	[1]
Molecular Formula	C ₅ H ₁₃ NO	[2]
Molecular Weight	103.16 g/mol	[2]
Appearance	Syrupy liquid	[1]
Odor	Fishy	[1]
Melting Point	< 25 °C	[3]
Boiling Point	Not definitively reported	[3]
Solubility	Soluble in water	[1]
pKa	Not definitively reported for the cation	
Stability	Unstable, decomposes to trimethylamine	[1]

Synthesis and Purification

Synthesis of Neurine

Neurine can be synthesized by the reaction of acetylene with aqueous trimethylamine.[1]

While a detailed industrial protocol is not publicly available, the general reaction is as follows:



A generalized laboratory-scale synthesis protocol would involve bubbling acetylene gas through a concentrated aqueous solution of trimethylamine under controlled temperature and

pressure. The reaction progress can be monitored by techniques such as thin-layer chromatography.

Purification of Neurine

The purification of **neurine** from the reaction mixture presents a challenge due to its instability. As a quaternary ammonium hydroxide, it can be purified using methods developed for this class of compounds. Ion-exchange chromatography is a suitable method.

Experimental Protocol: Purification by Ion-Exchange Chromatography

- **Resin Preparation:** A strong basic anion-exchange resin is converted to the hydroxide form by washing with a solution of a strong base, such as sodium hydroxide, followed by rinsing with deionized water until the eluent is neutral.
- **Sample Loading:** The crude **neurine** solution is loaded onto the prepared anion-exchange column.
- **Elution:** The column is eluted with deionized water. The **neurine** hydroxide is retained on the resin, while unreacted starting materials and byproducts are washed away.
- **Product Collection:** The purified **neurine** can then be eluted from the resin using a suitable counter-ion solution, or used directly as a resin-bound reagent depending on the subsequent application. Given **neurine**'s instability, in-situ use or immediate use after elution is recommended.

Characterization

Nuclear Magnetic Resonance (NMR) Spectroscopy

^1H and ^{13}C NMR spectroscopy are essential tools for the structural confirmation of **neurine**.

Expected ^1H NMR Spectral Data:

- **Vinyl Protons:** Three protons in the vinyl group ($-\text{CH}=\text{CH}_2$) would exhibit complex splitting patterns (multiplets) in the olefinic region of the spectrum (typically 5-7 ppm).

- Methyl Protons: Nine equivalent protons of the three methyl groups ($-\text{N}(\text{CH}_3)_3$) would appear as a sharp singlet in the upfield region (typically 3-4 ppm).

Expected ^{13}C NMR Spectral Data:

- Vinyl Carbons: Two distinct signals for the two carbons of the vinyl group.
- Methyl Carbons: A single signal for the three equivalent carbons of the methyl groups.

Sample Preparation for NMR:

- Dissolve a small amount of the purified **neurine** sample in a deuterated solvent (e.g., D_2O) to a concentration of 5-25 mg/mL for ^1H NMR and 50-100 mg/mL for ^{13}C NMR.
- Filter the solution to remove any particulate matter.
- Transfer the solution to an NMR tube.

Mass Spectrometry (MS)

Mass spectrometry can be used to determine the molecular weight and fragmentation pattern of **neurine**.

Expected Mass Spectrum:

- Molecular Ion: The molecular ion $[\text{CH}_2=\text{CHN}(\text{CH}_3)_3]^+$ would be observed at $m/z = 86$.
- Fragmentation: Common fragmentation pathways for quaternary amines include the loss of methyl groups.

Biological Activity and Neurotoxicity

Current research indicates that **neurine**'s primary biological role is that of a neurotoxin rather than a signaling molecule.^{[4][5]}

Mechanism of Neurotoxicity

The precise mechanism of **neurine**'s neurotoxicity is not fully elucidated but is thought to involve disruption of normal neuronal function.^[4] As a structural analog of choline, it may

interfere with cholinergic systems. General mechanisms of neurotoxicity often involve:

- **Receptor Interaction:** Interference with neurotransmitter receptors.
- **Ion Channel Disruption:** Altering the function of ion channels essential for neuronal signaling.
- **Enzyme Inhibition:** Inhibiting critical enzymes, such as acetylcholinesterase.[\[6\]](#)
- **Oxidative Stress and Mitochondrial Dysfunction:** Inducing cellular stress and impairing energy metabolism.[\[1\]](#)

Role in Alzheimer's Disease Pathology

Research has suggested a potential link between **neurine** and Alzheimer's disease. Studies have shown that **neurine** can elevate the levels of secreted amyloid-beta precursor protein (A β PP) and amyloid-beta (A β) peptides, and reduce neuronal cell viability in culture.[\[7\]](#) This suggests that **neurine**, as a breakdown product of acetylcholine, could contribute to the neurodegenerative processes seen in Alzheimer's disease.

Experimental Protocols for Studying Neurotoxicity

Cell Viability Assays

5.1.1. Lactate Dehydrogenase (LDH) Cytotoxicity Assay

This assay measures the release of LDH from damaged cells into the culture medium.[\[8\]](#)[\[9\]](#)

Protocol:

- **Cell Culture:** Plate neuronal cells (e.g., human neuroblastoma SK-N-SH cells) in a 96-well plate and culture until they reach the desired confluence.
- **Treatment:** Treat the cells with varying concentrations of **neurine** for a specified period (e.g., 24 hours). Include untreated and positive controls (e.g., a lysis buffer).
- **Sample Collection:** After incubation, carefully collect the cell culture supernatant.
- **LDH Measurement:** Add the supernatant to a new plate containing the LDH assay reagent mixture (containing lactate, NAD⁺, and a tetrazolium salt).

- Incubation and Reading: Incubate the plate at room temperature, protected from light, for approximately 30 minutes. Measure the absorbance at 490 nm using a microplate reader.
- Data Analysis: Calculate the percentage of cytotoxicity relative to the positive control.

5.1.2. MTS Cell Viability Assay

This colorimetric assay measures the reduction of a tetrazolium salt (MTS) by metabolically active cells.^{[10][11]}

Protocol:

- Cell Culture and Treatment: Follow the same procedure as for the LDH assay.
- MTS Reagent Addition: After the treatment period, add the MTS reagent directly to the wells containing the cells and media.
- Incubation: Incubate the plate for 1-4 hours at 37°C.
- Absorbance Reading: Measure the absorbance at 490 nm.
- Data Analysis: Express cell viability as a percentage of the untreated control.

Western Blot for Amyloid-beta Precursor Protein (A β PP)

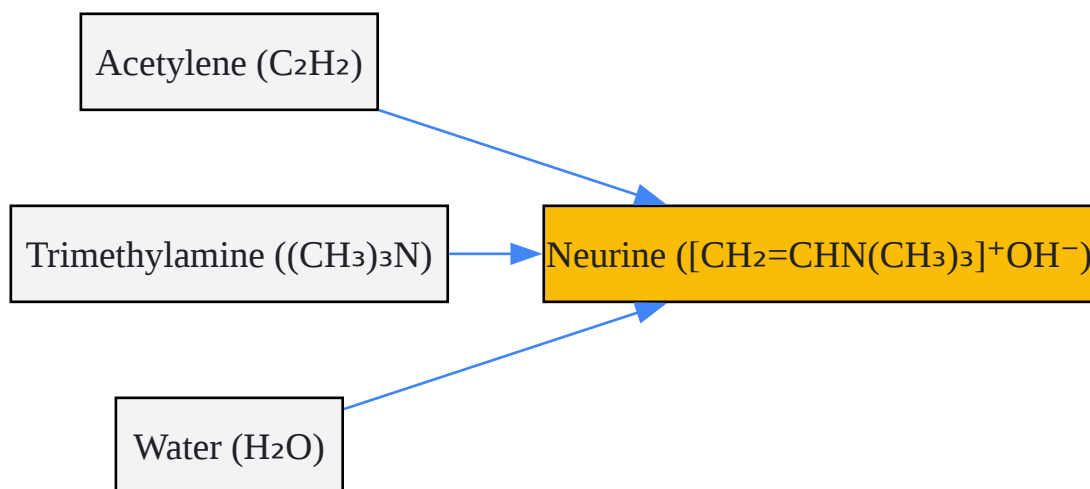
This technique can be used to quantify changes in A β PP levels in response to **neurine** treatment.^[12]

Protocol:

- Cell Culture and Treatment: Treat neuronal cells with sub-toxic concentrations of **neurine**.
- Protein Extraction: Lyse the cells and collect the protein lysate.
- Protein Quantification: Determine the protein concentration of each lysate.
- SDS-PAGE: Separate the proteins by size using sodium dodecyl sulfate-polyacrylamide gel electrophoresis.

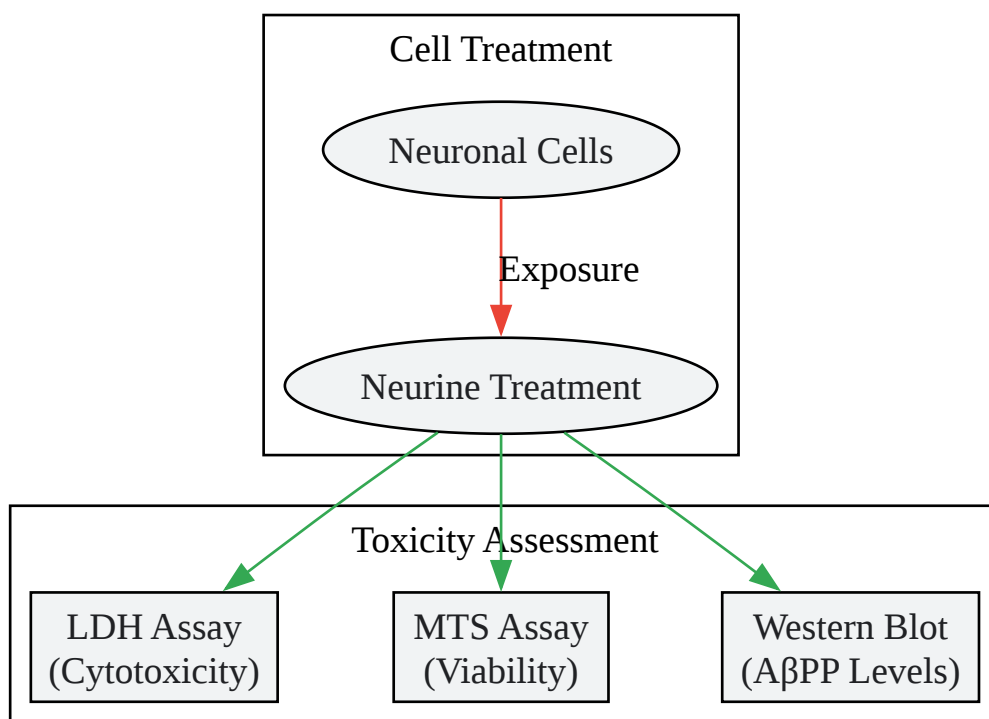
- Transfer: Transfer the separated proteins to a nitrocellulose or PVDF membrane.
- Blocking: Block the membrane to prevent non-specific antibody binding.
- Primary Antibody Incubation: Incubate the membrane with a primary antibody specific for A β PP.
- Secondary Antibody Incubation: Incubate with a horseradish peroxidase (HRP)-conjugated secondary antibody.
- Detection: Detect the protein bands using a chemiluminescent substrate and imaging system.
- Analysis: Quantify the band intensities relative to a loading control (e.g., β -actin or GAPDH).

Visualizations



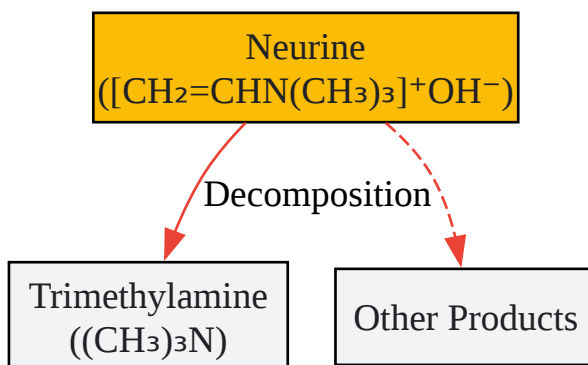
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Synthesis of **Neurine** from precursors.



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Experimental workflow for **neurine** toxicity assessment.



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Decomposition pathway of **Neurine**.

Conclusion

Neurine is a toxic quaternary ammonium compound with relevance to neurobiology and toxicology. While its physical and chemical properties are not exhaustively documented in

readily available literature, this guide provides a consolidated overview of the current knowledge. The provided experimental frameworks for synthesis, purification, characterization, and toxicological assessment offer a starting point for researchers investigating this compound. Future research should focus on elucidating the precise mechanisms of its neurotoxicity and its potential role in the pathology of neurodegenerative diseases. The lack of evidence for a direct role in signaling pathways suggests that its biological significance is primarily as a toxin.

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